

**D-11-Y**

Roll No.....

Total No. of Questions : 16]

[Total No. of Printed Pages : 4

**12<sup>th</sup>ARM(H/Z)JKUTL2024-25****611-Y****ZOOLOGY****(Biology)**

Time : 1½ Hours]

[Maximum Marks : 35

**SECTION-A****(OBJECTIVE TYPE QUESTIONS)**

1 each

1. Non-occurrence of Menses is named as ..... (Fill up)
  2. Expand the term 'VNTR'.
  3. Trichophyton and Epidermophyton are responsible for :  
(A) Elephantiasis  
(B) ✓ Ringworms  
(C) Typohoid  
(D) Pneumonia
- (Choose the correct one)

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Turn Over

( 2 )

4. *Atropa belladonna* has a haullucinogenic property. (True/False)
5. The term referred by Multinational Companies for the use of bio-resources is called :
- (A) Biopatent
- (B) Bioethics
- (C) ☒ Biopiracy
- (D) None of these
- (Choose the correct one)

**SECTION-B**

**(VERY SHORT ANSWER TYPE QUESTIONS)** 2 each

6. What do you mean by 'Blastocyst formation' ?
7. Write down the significance of Intrauterine Devices.

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**D-11-Y**

( 3 )

8. Write down *two* preventive measures of Amoebiasis.
9. Name *two* principles of Biotechnology.
10. Write down *two* applications of Gene Therapy.

**SECTION-C****(SHORT ANSWER TYPE QUESTIONS)**

3 each

11. Draw a neat diagram of Microscopic Anatomy of Testis.
12. Write down *three* preventive measures of AIDS.
13. Give *three* points of difference between Linkage and Crossing over. <https://www.jkboseonline.com>
14. What are the goals of 'Human Genome Project' ?
15. What are Vaccines ? Give their significance.

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**SECTION-D****(LONG ANSWER TYPE QUESTIONS)**

5

10. What do you mean by Sex-linked Inheritance ? Elaborate your answer with reference to Haemophilia and Color blindness.

*Or*

Define Evolution. Write down in brief Modern Synthetic theory of Evolution.

**B-11-X**

Roll No.....

Total No. of Questions : 16]

[Total No. of Printed Pages : 4

**12<sup>th</sup>ARM(SZ)JKUT2024****, 1111-X****ZOOLOGY****(Biology)****Time : 1½ Hours]****[Maximum Marks : 35*****General Instructions :***

This question paper contains four Sections :

(i) **Section-A** contains *five* Very Very Short Answer Type Questions of 1 mark each.  $1 \times 5 = 5$  marks

(ii) **Section-B** contains *five* Very Short Answer Type Questions of 2 marks each to be answered in **20-30** words.

 $2 \times 5 = 10$  marks

(iii) **Section-C** contains *five* Short Answer Type Questions of 3 marks each to be answered in **100-150** words.  $3 \times 5 = 15$  marks

(iv) **Section-D** contains *one* Long Answer Type Question to be answered in **150-200** words.  $1 \times 5 = 5$  marks

**12<sup>th</sup>ARM(SZ)JKUT2024-1111-X****B-11-X****Turn Over**

## SECTION-A

(VERY VERY SHORT ANSWER TYPE QUESTIONS) 1 each

1. Site of fertilization in mammals is :
- (A) Ovary
  - (B) Uterus
  - (C) Vagina
  - ☒ (D) Fallopian tube
2. Trisomic condition is represented as :
- ☒ (A)  $2n + 1$
  - (B)  $2n - 2$
  - (C)  $2n + 2$
  - (D) None of these
3. Inheritance is controlled by Multiple alleles in :
- (A) Colour-blindness
  - (B) Sickle cell anaemia
  - ☒ (C) Phenyl ketonuria
  - (D) Blood group

4. 'AZT' is used in treatment of :

(A) Malaria

~~(B)~~ AIDS

(C) T.B.

(D) Kala-Azar

5. The process of vaccination was discovered by :

~~(A)~~ Edward Jenner

(B) Robert Koch

(C) Louis Pasteur

(D) Von-Behring

### SECTION-B

(VERY SHORT ANSWER TYPE QUESTIONS)

2 each

6. What do you mean by 'Amniocentesis' ?

7. What is Placenta ? Write down its functions.

8. What are Darwin's finches ?
9. Write down preventive measures of 'Hepatitis'.
10. What do you mean by 'Biosafety Issues' ?

### SECTION-C

#### (SHORT ANSWER TYPE QUESTIONS)

3 each

11. Draw a well labelled diagram of Microscopic anatomy of testis.
12. Write a short note on colour blindness in human beings.
13. Write down the applications of Gene therapy.
14. Explain the term 'MTP' in brief.
15. Write a short note on drug abuse.

### SECTION-D

#### (LONG ANSWER TYPE QUESTIONS)

5

16. What do you mean by 'DNA Fingerprinting' ? Explain in brief 'Human Genome Project' and its significance.

Or

Explain Hardy-Weinberg's principle. Add a note on adaptive radiation.



**B-24-X**

Total No. of Questions 14]

[Total No. of Printed Pages 3

**XIIARJKUT23****9124-X****ZOOLOGY****(Biology)**

Time : 1½ Hours]

[Maximum Marks : 35

**SECTION-A****(OBJECTIVE TYPE QUESTIONS)****1 each**

1. Expand the term 'MTP'.
2. The last and only chromosome pair in humans is called 'Autosome'.

**(True/False)**

3. The causative agent of 'Filariasis' is .....

**(Fill in the blank)**

4. Name a volatile alkaloid present in Tobacco.

XIIARJKUT23-9124-X

**B-24-X****Turn Over**

**SECTION-B****(VERY SHORT ANSWER TYPE QUESTIONS)** 2 each

5. What are the points of difference between 'GIFT' and 'ZIFT' ?
6. Enlist the symptoms of Typhoid.
7. What is the function of Endonucleases ?

*Or*

Give the application of genetically Engineered Insulin. ,

**SECTION-C****(SHORT ANSWER TYPE QUESTIONS)** 3 each

8. Draw a well labelled diagram of a T.S. of a Human Ovary.
9. What are the steps involved in DNA Fingerprinting ?
10. Explain briefly Hardy-Weinberg equilibrium.
11. What is a Parasites ? Add a note on Ascariasis in humans.
12. What are 'Bio-safety issues' ?

*Or*

Give an account of Gene-cloning.

**SECTION-D****(LONG ANSWER TYPE QUESTIONS)****5 each**

13. What is Sexual Reproduction ? Describe the process of gamete formation in human males.

*Or*

Define Uni-parental Reproduction. Explain any *three* modes of asexual reproduction in the animal world.

14. Explain the various Mendelian disorders found in humans.

*Or*

Discuss the sex-determination mechanisms found in birds and honeybees.

**F-4-A**

Roll No.....

Total No. of Questions : ~~14~~+11]

[Total No. of Printed Pages : 8

**XIIAPBIKD22****6904-A****ZOOLOGY****(Biology)****(New/Old Course)****Time : 1½ Hours]****[Maximum Marks : 35**

**NOTE** :— The questions in the question paper are based on revised and pre-revised syllabus marked as “New Course” and “Old Course” respectively and candidates are advised to appear in the relevant course meant for them. Candidates who may attempt the questions partly from “**New Course**” and partly from “**Old Course**” will not be awarded. Candidates are also advised to record “**New Course**” or “**Old Course**” as the case may be, on the front page of the answer-book.

XIIAPBIKD22—6904-A (New)

Turn Over

**F-4-A**

**(New Course)****Section-A****(Objective Type Questions/Do as directed)**

1 each

1. Morphologically and genetically similar individuals produced as a result of asexual reproduction are known as .....  
(Fill in the blank)
2. The probability of sons being colourblind from a normal father and a colourblind mother is :  
(A) 25%  
(B) 50%  
(C) 75%  
(D) 100%  
(Choose the correct one)
3. Name the causative agent of Typhoid.
4. HIV stands for ?

**Section-B****(Very Short Answer Type Questions)**

2 each

5. Write down the functions of luteinizing hormone in human males and females.
6. Write short note on Vaccines.
7. Define recombinant DNA technology.

*Or*

Name a few important products of Biotechnology.

XIIAPBIKD22-6904-A (New)

**F-4-A**

( 3 )

**Section-C****(Short Answer Type Questions)****3 each**

8. Draw a labelled diagram of saggital section of human female breast.
9. Explain the sex determination in honey bee.
10. What are Darwin's Finches ?
11. Differentiate between cell mediated and humoral immunity.
12. Write short note on Human Insulin.

*Or*

Discuss Bio-safety issues.

**Section-D****(Long Answer Type Questions)****5 each**

13. Define Gametogenesis. Explain in brief the various stages of spermatogenesis with diagrammatic representation.

*Or*

Describe the hormonal control of menstrual cycle in human females.

14. Why human genome is called as Mega Project ? Write down its goals and future implications.

*Or*

Describe the Hardy-Weinberg Principle. What are the conditions that maintain the Hardy-Weinberg Equilibrium ?

**F-4-A**

( 5 )

**XIIAPBIKD22****6904-A****ZOOLOGY****(Biology)****(New/Old Course)****Time : 1½ Hours]****[Maximum Marks : 35**

**NOTE :—** The questions in the question paper are based on revised and pre-revised syllabus marked as “New Course” and “Old Course” respectively and candidates are advised to appear in the relevant course meant for them. Candidates who may attempt the questions partly from “**New Course**” and partly from “**Old Course**” will not be awarded. Candidates are also advised to record “**New Course**” or “**Old Course**” as the case may be, on the front page of the answer-book.

XIIAPBIKD22—6904-A (Old)

Turn Over

**F-4-A**



***(Old Course)***

**(Long Answer Type Questions)**

**5 each**

1. Draw a labelled sketch of Human male reproductive system. How is oogenesis markedly different from spermatogenesis ?

*Or*

Name at least *five* important STD's (Sexually Transmitted Disease) and common techniques for their detection.

2. Define Hardy-Weinberg Principle. Explain at least *three* basic processes which change Hardy-Weinberg equilibrium and bring about variations at the genetic level.

*Or*

Name the chromosomal complement and effects of Down's syndrome and Patau's syndrome.

PBIKD22—6904-A (Old)

**1-A**

**(Short Answer Type Questions)**

3 each

3. Name the various barrier methods of contraception. Which one is most commonly used ?

4. State two points of difference between the mechanism of sex-termination in Birds and Humans.

5. Write down at least *three* points of difference between Non-specific and Specific Immunity. <https://www.jkboseonline.com>

6. Write a short note on Restriction Endonucleases.

**(Very Short Answer Type Questions)**

2 each

7. Write down the role of Sertoli Cells.

8. What are Vestigial Organs ? Give examples.

9. What is the significance of Vaccines ?

10. What do you mean by Cloning Vectors ?

**(MCQ/Objective Type Questions/Do as Directed)** 1 each

11. (i) Human Ovum is 'Alecithal'. (True/False)

(ii) Weismann proposed the theory of ..... of germplasm.

(Fill in the blank)

(iii) Trisomy of 21st Chromosome causes :

(A) Klinefelter's syndrome

(B) Turner's syndrome

(C) Sickle cell anaemia

(D) Down syndrome (Choose the correct one)

(iv) Cancer of Epithelial cells is Carcinoma. (Agree/Disagree )

(v) Which of the following is celebrated as 'No Tobacco Day' ?

(A) May, 29

(B) March, 23

(C) July, 21

(D) May, 31 (Choose the correct one)

**B-4-A**

Roll No.....

Total No. of Questions : 14]

[Total No. of Printed Pages : 3

**XIIKDAR21****5004-A****ZOOLOGY****(Biology)**

Time : 1½ Hours]

[Maximum Marks : 35

**Section-A****(Objective Type Questions)**

1 each

1. The Asexual reproduction is also called 'Somatogenic reproduction'.

(Yes/No)

2. Which of the following methods of sex-determination operates in Butterflies and Moths ?

(A) ZW - ZZ type

~~(B)~~ XX - XO type

(C) ZO - ZZ type

(D) None of these

(Choose the correct one)

XIIKDAR21—5004-A

Turn Over

**B-4-A**

( 2 )

3. Recovery of product, its refining and processing is called .....  
processing. (Fill-up)
4. 'Dolly' is not a 'GMO'. (True/False)

**Section-B****(Very Short Answer Type Questions)**

2 each

5. What do you mean by Assisted reproductive technologies ?
6. What do you mean by 'Cell mediated immunity' ?
7. Name the causative organism of 'Typhoid'. Which test is prescribed for detection of Typhoid.

*Or*

Write down the significance of Dairy farming.

**Section-C****(Short Answer Type Questions)**

3 each

8. Write a short note on prevention of STD's (Sexually Transmitted Diseases).
9. Write a short note on 'Australopithecus'.
10. Write down the significance of Modern Synthetic Theory of Evolution.

XIIKDAR21—5004—A

**B-4-A**

( 3 )

11. Define Sericulture. Write down the economic importance of Silk-worm.
12. Define Cloning. Support your answer with examples.

*Or*

**What is Gene therapy ? Write down its significance.**

**Section-D**

**(Long Answer Type Questions)**

**5 each**

13. Define Fertilization. Enlist the events upto the development of Blastocyst.

*Or*

**Explain in brief the need for Reproductive health. Also add a note on MTP (Medical Termination of Pregnancy).**

14. Define DNA fingerprinting. Write down in brief the applications of DNA fingerprinting.

*Or*

**Why is human genome project called a Mega project ? Write down its future implications.**

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